[INCH-POUND]
A-A-59401
24 September 1999
SUPERSEDING
MIL-G-80145B
22 February 1989

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINES, SURFACE, RECIPROCATING TABLE, HORIZONTAL SPINDLE, HAND FEED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers floor-mounted, hand-feed, horizontal-spindle, reciprocating-table, surface grinding machines. The grinding machines are intended for use in tool and die, maintenance, and production shops for rough and precision-finishing or form grinding operations.

2. CLASSIFICATION. The grinding machines shall be of the following sizes, shown in table I. The size to be furnished shall be as specified (see 7.2(b)). Machine sizes are shown in table I.

TABLE I. Machine sizes.

Size	Worktable		Minimum work height capacity, inches
	Width, inches	Length, inches	
612	6	12	11
614		14	
618		18	
715	7	15	14
818	8	18	

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3415

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall be a casting, fabrication, or component assembly constructed as a stable platform that supports all necessary components of the machine with minimal elongation, deflection, and vibration. A means shall be provided for leveling and anchoring the machine to the floor.

3.2.2 Saddle. The saddle shall provide crossfeed for the worktable by traveling along the Z-axis that is parallel to the axis of rotation of the grinding wheel spindle. Machineways built into the saddle shall provide the means for the worktable to travel longitudinally along the X-axis perpendicular to the Z-axis.

3.2.3 Worktable. The worktable shall be in accordance with the dimensions shown in table I for the machine size provided. Unless otherwise specified (see 7.2(c)), the worktable shall be equipped with the manufacturer's standard T-slots, threaded screw, or other mountings for work-holding fixtures and/or magnetic chucks provided.

3.2.4 Magnetic chuck. If required, a permanent magnetic or electromagnetic chuck shall be provided as specified (see 7.2(d)). Unless otherwise specified (see 7.2(e)), the working surface area of the chuck shall be the same as that of the worktable. The chuck shall be sealed for protection from coolant and grinding swarf.

3.2.5 Wheelhead column. A vertical column shall be provided for the support and transport of the wheelhead assembly. The column may be formed as part of or rigidly bolted to the bed.

3.2.6 Machineways. Machineways shall be protected from chips, dust, coolant, and other contaminants.

3.2.7 Wheelhead assembly. As a minimum, the wheelhead assembly shall include a housing and supporting frame, a grinding wheel spindle, a spindle drive system, and a grinding wheel guard.

3.2.8 Spindle motor. Unless otherwise specified (see 7.2(f)), the spindle motor size and speed shall be that considered by the manufacturer to be standard.

3.2.9 Grinding wheel. Unless otherwise specified (see 7.2(g)), the grinding wheel type and size shall be that considered standard by the manufacturer.

3.2.10 Controls. All controls shall be mounted at a location convenient to the operator and positioned to reduce operator fatigue. Unless otherwise specified (see 7.2(h)), the following controls, as a minimum, shall be provided:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

a. Emergency stop
b. Handwheels for position and feed on all axes
c. On/off spindle motor control, and when applicable, speed
d. On/off power and flow rate for the coolant supply
e. On/off power for the dust extractor

3.2.10.1 Control interlock. The controls for the coolant-circulating and/or dust-extractor systems shall be interlocked to provide only independent operation and prevent simultaneous operation of these systems. When a magnetic chuck is provided, the controls for the grinding wheel and chuck shall be interlocked to provide independent activation of the chuck and prevent operation of the grinding wheel without activation of the chuck.

3.2.10.2 Handwheels. All linear motion of the worktable and grinding wheel shall be provided by handwheels. Handwheels shall be provided with scales contiguously calibrated over the full range of motion. Unless otherwise specified (see 7.2(i)), the scales used for fine feed shall indicate linear motion in increments of not greater than 0.0001 inch.

3.2.10.3 Digital readout. When specified (see 7.2(j)), a digital readout system shall be provided. The system shall include all feedback devices and components necessary to provide a digital display for the full working range of the Y- and Z-axes. The display shall have a minimum resolution of six digits per axis. Unless otherwise specified (see 7.2(k)), system accuracy shall be at least ±0.00005 inch and repeatability shall be within 0.0001 inch measured over the full travel of each axis. The system shall selectively display in either the US or SI measurement system. All components of the digital readout system shall be protected from damage caused by the normal grinding environment of vibration, coolant splash, grinding swarf or dust, and electric power fluctuations.

3.2.11 Dust Extractor system. Unless otherwise specified (see 7.2(l)), a dust extractor system shall be provided for use during dry grinding.

3.2.12 Lubrication system. A lubrication system shall be provided to ensure adequate lubrication of all moving parts. When specified (see 7.2(m)), the lubrication system shall be completely automatic and shall be provided with interlocks to prevent machine operation in case of pressure loss. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing.

3.2.13 Coolant system. Unless otherwise specified (see 7.2(n)), a coolant system shall be provided for wet grinding with all necessary equipment and sufficient reservoir capacity to permit a continuous full flow of clean coolant throughout the work zone. The system shall have a means for draining and cleaning the reservoir, baffles, strainers, and filters. A means shall be provided for flushing the catch pans, troughs, and other areas where swarf may accumulate.

3.2.14 Electrical system. The electrical system shall conform to the American National Standards Institute /National Fire Protection Association (ANSI/NFPA) 79, "Electrical Standard for Industrial Machinery". Unless otherwise specified in the acquisition order (see 7.2(o)), the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

machine shall operate from a dual 230/460-volt, 3-phase, 60-hertz circuit and shall be wired for the voltage specified. The control system shall operate on a 120-volt, single-phase, 60-hertz circuit from the machine main power supply.

3.2.15 Electric motors. The grinding wheel, coolant pump, and dust extractor drive motor(s) shall have continuously lubricated bearings or sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1, "Motors and Generators". Unless otherwise specified (see 7.2(p)), the motors shall have dripproof enclosures.

3.2.16 Optional accessories. Optional accessories shall be provided as specified (see 7.2(q)).

3.3 Performance. Unless otherwise specified (see 7.2(r)), the grinding machine shall meet the alignment requirements shown in table II and the performance requirements shown in table III when tested in accordance with section 5.3 through 5.3.5.

TABLE II. Alignment requirements.

Parameter	Tolerance
Worktable flatness: deviation from mean	±0.0001 inch
Worktable travel: rise or fall, over full travel	±0.0001 inch
Wheel spindle, peripheral runout	0.0002 inch TIR[1]
Wheel spindle, axial slip	0.0001 inch TIR
Squareness of axes: X- and Z-axes shall be square to each other, and the Y-axis shall be square to the plain described by X and Z	0.001 inch/foot
Machineway pairs: surfaces flat and straight, edges parallel	0.0001 inch/foot

[1] TIR = Total indicator readout.

TABLE III. Performance requirements.

Parameter	Tolerance
Workpiece: surface flatness	0.0001 inch/foot
Workpiece: surface parallel	0.0001 inch/foot
Workpiece: surface finish	0.000008 inch
Workpiece: finished dimensions	±0.0001 inch

3.4 Machine dimensions and weight. The maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(s)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.9, "Safety Requirements for the construction, Care, and Use, of Grinding Machines", or an equivalent standard from the country of manufacture. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "General Industry, Safety and Health Standards". If a conflict arises between ANSI B11.9 or its equivalent and the OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(t)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(u)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(v)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Machine model designation
d. Power input (volts, total amperes, phase, frequency)
e. Short-circuit/over-current rating
f. Contract number or order number
g. National stock number or plant equipment code
h. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(w)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(x)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(y)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(z)). Unless otherwise specified (see 7.2(aa)), the machine shall be tested in accordance with environmental conditions of 40 to 104 degrees Fahrenheit and 20 to 95 percent humidity.

5.3.2 Operational test. The grinding machine shall be operated in accordance with the manufacturer's standard operating procedures. The grinding machine shall be operated at no load for not less than 1 hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Alignment tests. After the operational test, the alignment shall be verified and shall meet the requirements shown in table II. Alignment shall be verified again following the performance tests.

5.3.4 Performance tests. The performance tests shall be conducted in accordance with the manufacturer's standard operating procedure. The grinding machine shall be tested by grinding the top and bottom surfaces of five test blocks, with one block located at each corner and one block in the center of the worktable. The blocks shall have top and bottom surfaces of at least 4 square inches and be at least 1 inch thick. The grinding procedure shall remove a minimum of 0.001 inch from both surfaces. The finished surfaces of the blocks shall be flat, parallel, and have a surface roughness within the requirements shown in table III. The blocks shall be of the same dimensional thickness within 0.0001 inch per foot of separation. Four roughness samples shall be taken on each surface. The lay of the roughness assessment line shall be perpendicular in two of the four samples and parallel in the other two samples to the scoring of the surface measured. Total surface roughness shall be defined as the measurement of the highest peak to the lowest valley in an assessed length of 0.1 inch.

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(bb)).

5.4 Acceptance. Unless otherwise specified (see 7.2(cc)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with tables II and III. Failure of the machine to meet the performance requirements shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the products shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(dd)).

7. NOTES

7.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document
b. Size of grinding machine required (see 2)
c. Worktable mounts, if different (see 3.2.3)
d. Magnetic chuck, as specified (see 3.2.4)
e. Magnetic chuck surface area, if different (see 3.2.4)
f. Spindle motor size and speed, if different (see 3.2.8)
g. Grinding wheel type and size, if different (see 3.2.9)
h. Controls, if different (see 3.2.10)
i. Handwheel scales, if different (see 3.2.10.2)
j. Digital readout, if required (see 3.2.10.3)
k. Digital readout specification, if different (see 3.2.10.3)
l. Dust extractor, if different (see 3.2.11)
m. Automatic lubrication, if required (see 3.2.12)
n. Coolant system, if different (see 3.2.13)
o. Voltage, if different (see 3.2.14)
p. Motor enclosure, if different (see 3.2.15)
q. Optional accessories, as specified (see 3.2.16)
r. Performance, if different (see 3.3)
s. Machine dimensions and weight, as specified (see 3.4)
t. System of units required (see 3.6)
u. Dual system of units, if required (see 3.6)
v. Nameplate, if different (see 3.7)
w. Lubrication plate or chart, if required (see 3.8)
x. Testing, if different (see 5.3)
y. Test responsibility, if different (see 5.3.1)
z. Certification of calibration, as specified (see 5.3.1)
aa. Test environmental conditions, if different (see 5.3.1)
bb. Optional tests, as specified (see 5.3.4)
cc. Acceptance, if different (see 5.4)
dd. Packaging requirements, as specified (see 6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

7.2 Sources of documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

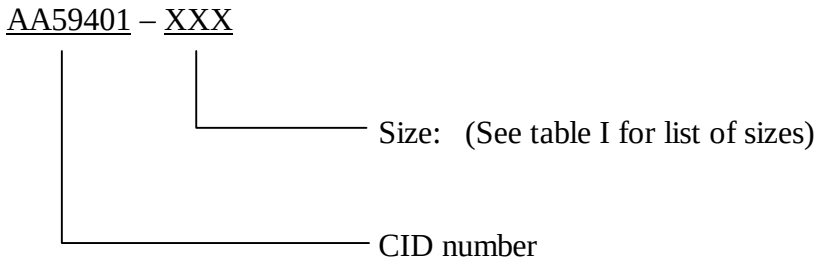
7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.4 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.5 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.3 Part identification number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Army - AR, SM
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3415-0251)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.